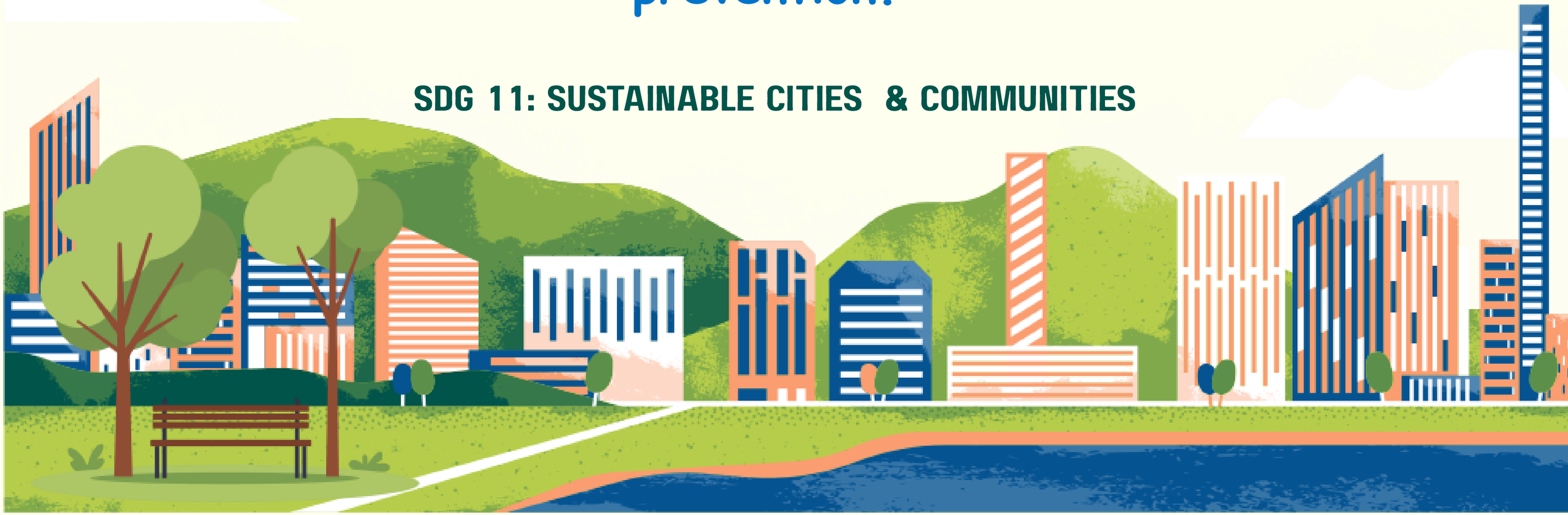
Cracks to Craters

Topic - Awareness in worsening road conditions and prevention.

SDG 11: SUSTAINABLE CITIES & COMMUNITIES



INTRODUCTION

Worsening road conditions, especially potholes, are a major challenge in urban areas, causing accidents, vehicle damage, and traffic issues.

These problems often go unnoticed until they become severe. They also increase travel time and create inconvenience for daily commuters.

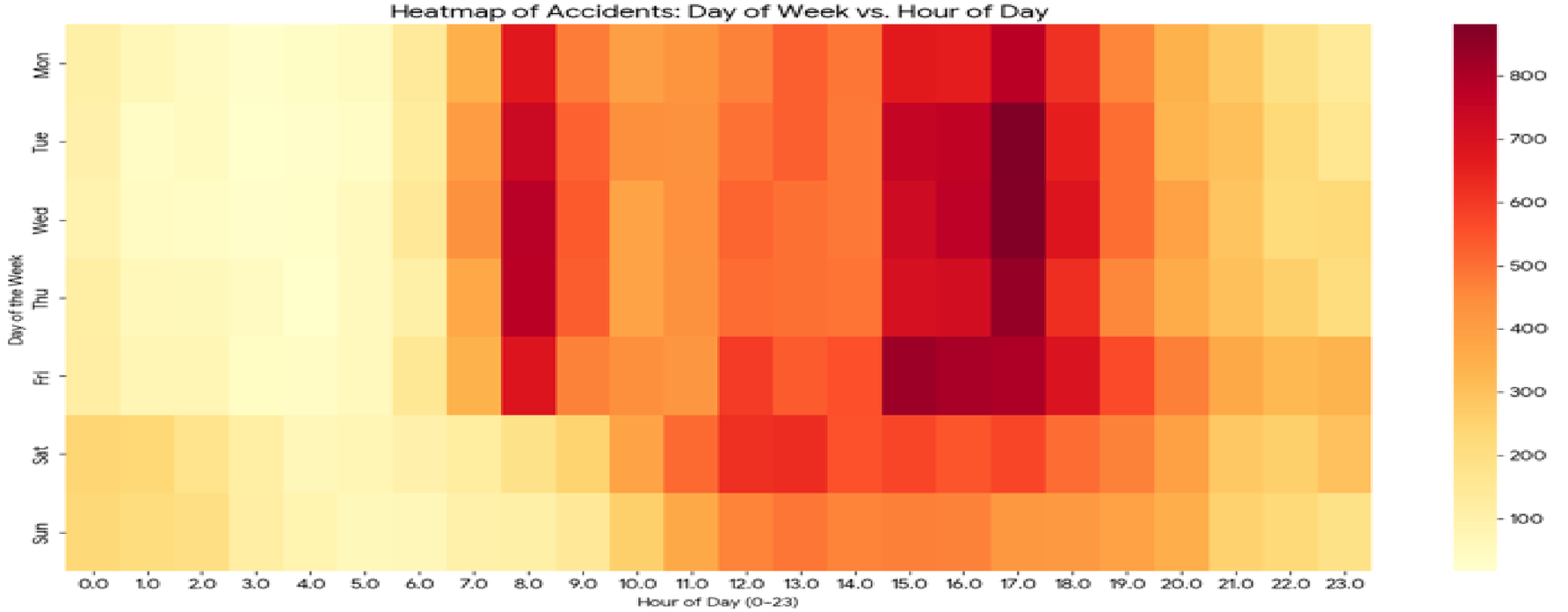
This project uses Artificial Intelligence (AI) to detect potholes, raise awareness, and enable faster action by authorities. It is based on SDG 11 to improve road safety and support Sustainable Cities.

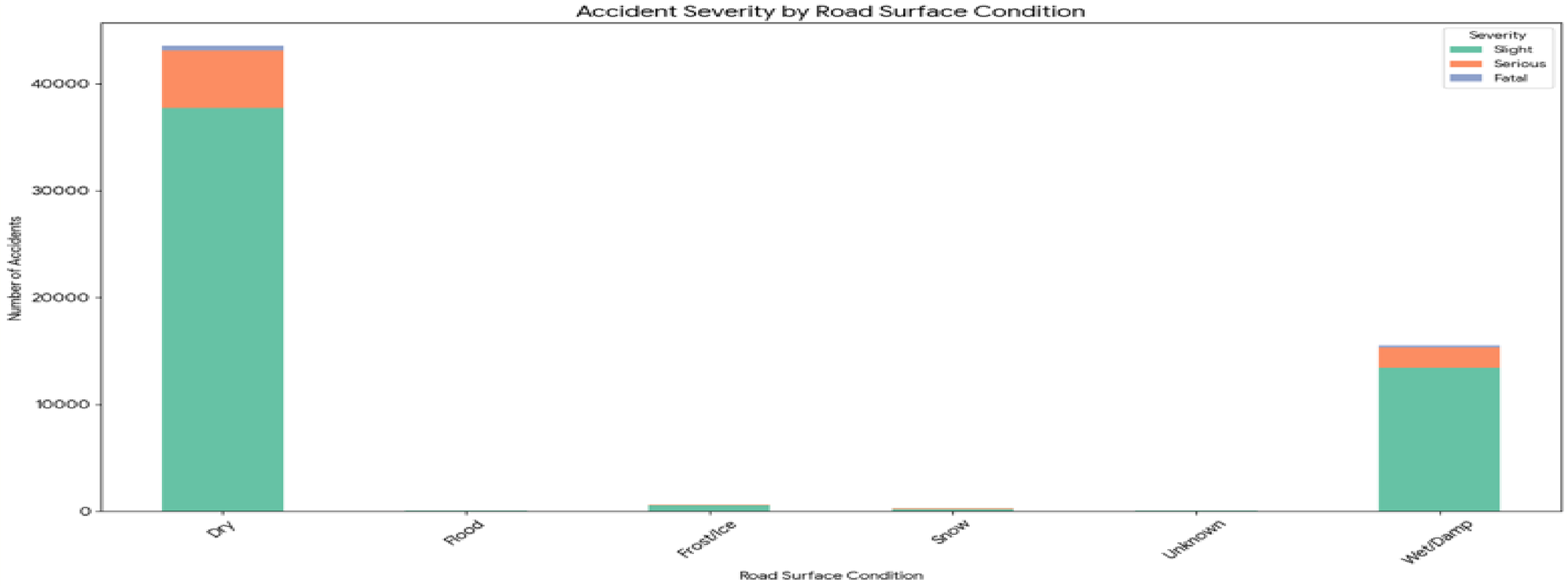
Causes of Potholes

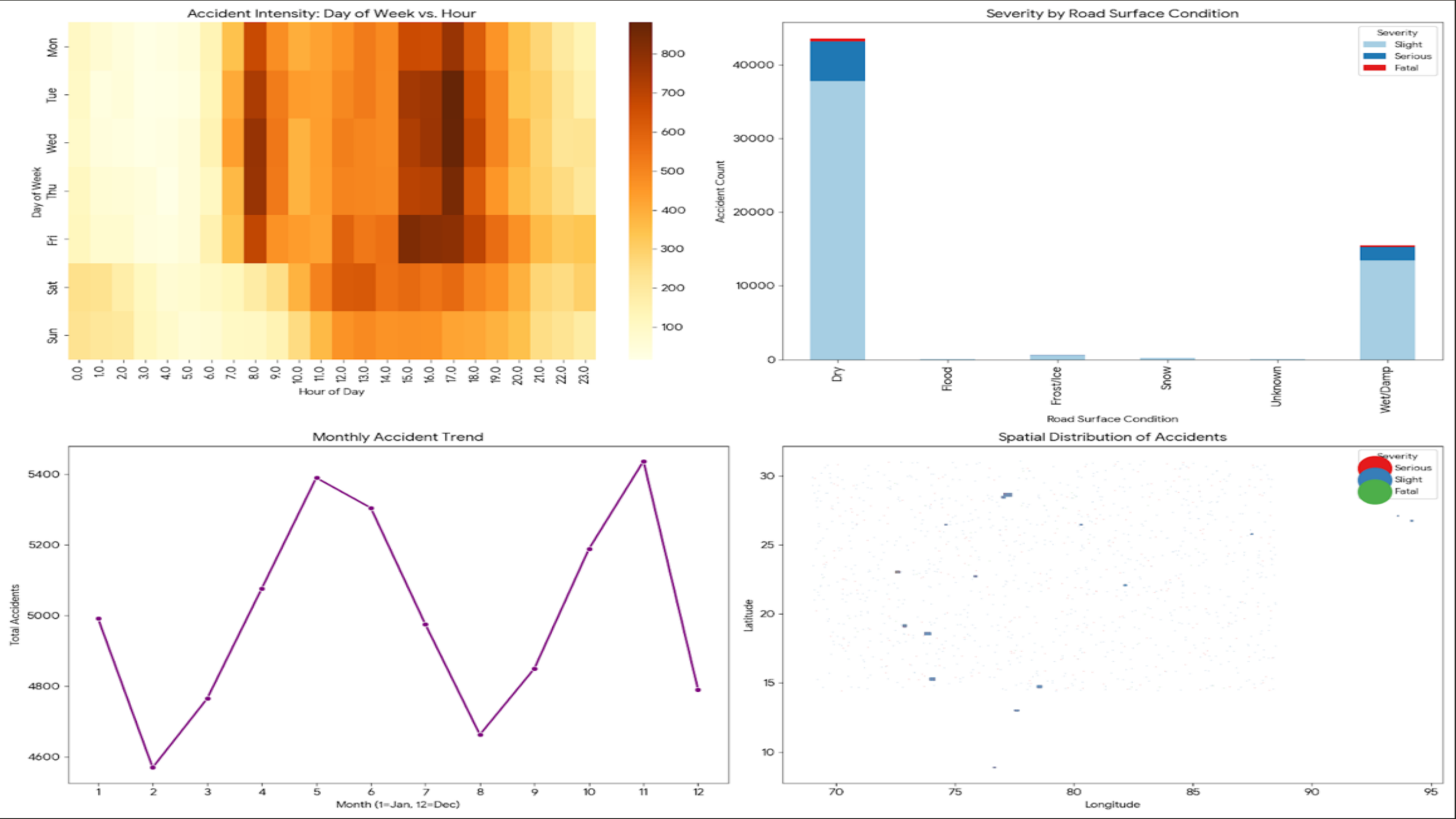
- Water seepage: Rainwater enters cracks and weakens the road base.
- Heavy traffic: Continuous vehicle load causes surface damage.
- Poor construction: Low-quality materials reduce road durability.
- Temperature variation: Expansion and contraction create cracks.
- Lack of maintenance: Small damages are not repaired on time .
- Aging roads: Old roads naturally deteriorate over time.

Impact

- Increase risk of accidents and injuries.
- Cause damage to tires and vehicles.
- Increase traffic and create delays.
- Increase fuel usage and repair costs.
- Make roads unsafe, especially in rain or at night.
- Affect emergency services response time.
- Cause discomfort during travel.
- Increase pollution due to slow-moving traffic.
- Impact local businesses due to poor connectivity.


Heatmap of Accidents: Day of Week vs. Hour of Day
Day of the Week
Mon
Tue
Wed
Thu
Fri
Sat
Sun
0.0 1.0 2.0 3.0 4.0 5.0 6.0 7.0 8.0 9.0 10.0 11.0 12.0 13.0 14.0 15.0 16.0 17.0 18.0 19.0 20.0 21.0 22.0 23.0
Hour of Day (0-23)
800
700
600
500
400
300
200
100


Accident Severity by Road Surface Condition
Severity
Slight
Serious
Fatal
Number of Accidents
40000
30000
20000
10000
0
Dry
Road
Frost/Ice
Snow
Unknown
Wet/Damp
Road Surface Condition


Accident Intensity: Day of Week vs. Hour
Day of Week
Mon
Tue
Wed
Thu
Fri
Sat
Sun
Hour of Day
0.0 1.0 2.0 3.0 4.0 5.0 6.0 7.0 8.0 9.0 10.0 11.0 12.0 13.0 14.0 15.0 16.0 17.0 18.0 19.0 20.0 21.0 22.0 23.0
100 200 300 400 500 600 700 800
Severity by Road Surface Condition
Severity
Slight
Serious
Fatal
Accident Count
0 10000 20000 30000 40000
Dry
Flood
Frost/Ice
Snow
Unknown
Wet/Damp
Road Surface Condition
Monthly Accident Trend
Total Accidents
4600 4800 5000 5200 5400
1 2 3 4 5 6 7 8 9 10 11 12
Month (1=Jan, 12=Dec)
Spatial Distribution of Accidents
Severity
Serious
Slight
Fatal
Latitude
10 15 20 25 30
Longitude
70 75 80 85 90 95

Empathy map

Says

- Roads are in bad condition.
- This is dangerous to drive on.
- Repairs are always delayed.
- This damages my vehicle.

Thinks

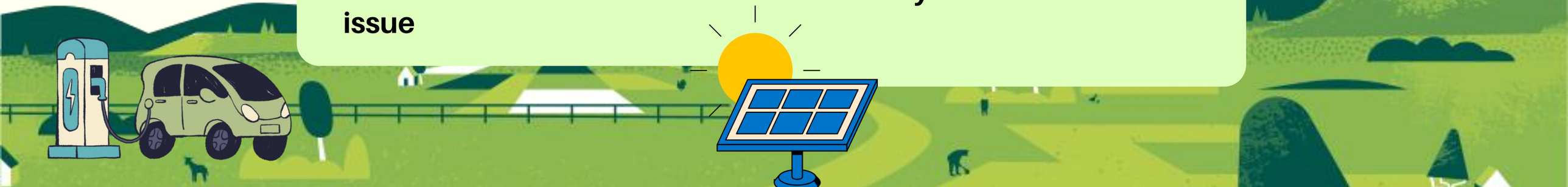
- Someone might get into an accident.
- Why isn't this fixed yet?
- Is there a safer route?
- This will cost me money to keep repairing my vehicle

Does

- Avoids damaged roads
- Drives slowly or carefully
- Changes routes frequently
- Complains or reports the issue

Feels

- Frustrated
- Unsafe
- Worried
- Annoyed



5W1H

What

What is the problem? -

Increasing potholes on roads

What causes it?

Water damage, traffic load, poor construction

What is affected?

Safety, vehicles, and road quality

Why

Why do potholes form?

Due to water seepage and weak materials

Why is it a concern?

It leads to accidents and damage

Why is it not fixed quickly?

Due to delayed maintenance

Who

Who is affected?

Drivers, commuters, pedestrians

Who is responsible?

Local authorities and contractors

Who can help?

Citizens by reporting issues

Where

Where do potholes occur?

Roads, highways, streets

Where are they common?

High-traffic and poorly maintained areas

Where should they be reported?

Local municipal platforms

When

When do potholes form?

Mostly during and after rainy seasons

When do they worsen?

With continuous traffic and neglect

When should they be fixed?

As soon as detected

How

How can it be solved?

Using AI detection and timely repair

How can people help?

By reporting and spreading awareness

How can damage be reduced?

Through proper maintenance and materials

PROTOTYPE

Our prototype transforms a simple camera feed into a powerful safety engine by seamlessly blending cutting-edge AI with real-world impact.

Powered by a YOLOv8 Nano model, it acts as a digital eye—scanning roads in real time, detecting potholes with remarkable speed and precision, and converting raw visual data into meaningful insights.

Each detection is more than just a box on the screen; it represents a potential hazard identified, quantified, and ready to be acted upon.

This bridges the gap between advanced technology and everyday safety, turning pixels into proactive protection.

What elevates this prototype is its ability to think ahead. By integrating detection data with predictive modeling, it begins to forecast accident-prone zones—shifting the approach from reactive repairs to preventive action.

Even in its current form, the prototype reflects a scalable vision for smarter cities—where roads are continuously monitored, risks are anticipated, and decisions are driven by data.

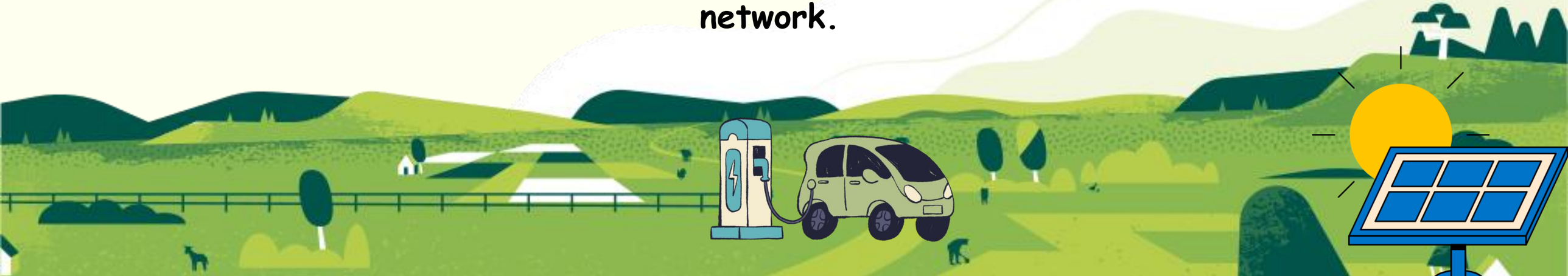
It's not just a model; it's a glimpse into a future where technology actively safeguards human lives and reshapes urban mobility.

FUTURE SCOPE

This project has the potential to evolve into a powerful, city-wide smart infrastructure system by integrating with CCTV networks, vehicle cameras, and mobile devices for real-time pothole detection.

With continuous data collection and GIS mapping, authorities can visualize road conditions, prioritize repairs intelligently, and allocate resources more efficiently

. The system can further expand into a public mobile application, enabling citizens to receive live alerts and even contribute data, creating a collaborative and responsive urban safety network.



On a more advanced level, the project can incorporate sophisticated machine learning models to predict accident-prone zones before incidents occur, shifting from reactive to preventive road safety.

Its scalability and real-world impact make it a promising solution for smart city initiatives, with strong potential for collaboration with government bodies and mobility startups.

Ultimately, this project envisions transforming urban transport into a safer, data-driven ecosystem, aligning technology with the future of sustainable living.

CONCLUSION

Potholes are a major issue affecting road safety, vehicles, and daily life.

Timely detection and proper maintenance are essential to prevent further damage.

AI-based solutions can enable real-time pothole detection, accurate reporting, and faster response by authorities.

They also help in predicting road damage and improving maintenance planning, leading to safer, smarter, and more reliable roads for everyone.

THANK YOU !

-Ronak Pal ,Osh Singh, Kota Katyayani Sri